



**The University of Azad Jammu & Kashmir,
Muzaffarabad**

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Course Title: Database System

Course Code: CS-2204

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Lab: 10

Department Of Software Engineering

✓ **Scalar Function**

1. Introduction

SQL functions are built-in operations used to perform specific tasks on database values. A **scalar function** works on one row at a time and returns one value for each row.

This lab focuses on **String Functions** in MySQL. String functions are used to clean, format, combine, search, and extract text data.

In this lab, the Customer and Product tables are used. The database contains some intentionally messy data such as extra spaces, uppercase and lowercase names, and NULL values so that different string functions can be practiced.

2. Objectives

The objectives of Part A are:

1. To understand scalar SQL functions.
2. To use string functions in MySQL.
3. To remove unnecessary spaces from text.
4. To convert text into uppercase and lowercase.
5. To count characters in a string.
6. To combine multiple strings.
7. To extract usernames and domains from email addresses.
8. To extract specific characters from names.
9. To mask phone numbers.
10. To replace spaces with other characters.
11. To pad numeric IDs with leading zeros.
12. To locate specific text inside product names.

3. Database and Tables

Database Name

scalar_lab

Tables

The lab uses two tables:

- Customer
- Product

Customer Table

Column	Description
CustID	Customer ID
CustName	Customer name
Email	Customer email
City	Customer city
Phone	Customer phone
JoinDate	Customer joining date
DOB	Customer date of birth

Product Table

Column	Description
ProdID	Product ID
ProdName	Product name
Category	Product category
Price	Product price
StockQty	Available stock
LaunchDate	Product launch date

The lab manual defines these two tables and their columns in the schema section.

4. String Functions Used

Trim(), Locate () etc

5. Part A — Lab Tasks

Task A1 — Remove Leading and Trailing Spaces

SQL Query

```
SELECT CustID,CustName AS Original,  
       TRIM(CustName) AS CleanedName  
FROM Customer;
```

Extra options				CustID	Original	CleanedName
☐	 Edit	 Copy	 Delete	1	Ali Khan	Ali Khan
☐	 Edit	 Copy	 Delete	2	Sara Iqbal	Sara Iqbal
☐	 Edit	 Copy	 Delete	3	HAMZA RAZA	HAMZA RAZA
☐	 Edit	 Copy	 Delete	4	Ayesha Noor	Ayesha Noor
☐	 Edit	 Copy	 Delete	5	bilal ahmed	bilal ahmed
☐	 Edit	 Copy	 Delete	6	Fatima Sheikh	Fatima Sheikh
☐	 Edit	 Copy	 Delete	7	Usman Tariq	Usman Tariq
☐	 Edit	 Copy	 Delete	8	Maira Javed	Maira Javed
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Task A2 — Convert Names to Uppercase and Lowercase

➤ Objective

Display every customer name in uppercase and lowercase.

➤ SQL Query

```
SELECT
    CustID,
    UPPER(CustName) AS UpperName,
    LOWER(CustName) AS LowerName
FROM Customer;
```

➤ Explanation

UPPER() converts the name to uppercase, while LOWER() converts it to lowercase.

For example:

HAMZA RAZA

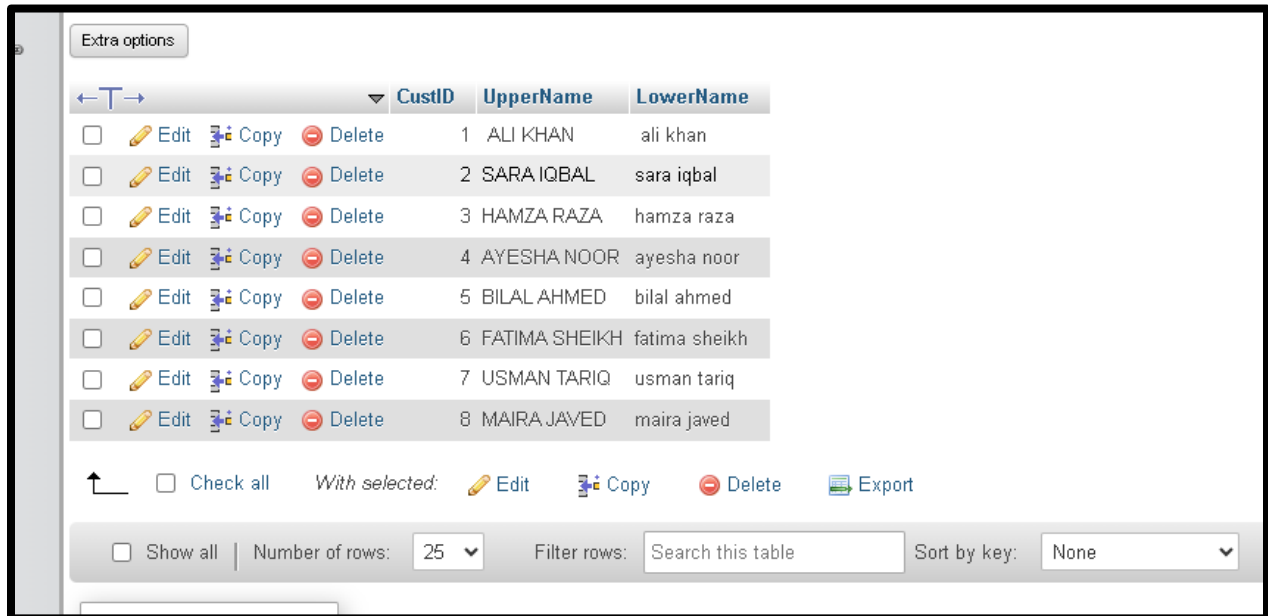
can be displayed in lowercase as:

hamza raza

➤ Result

The query displays:

- Customer ID
- Uppercase name
- Lowercase name



				CustID	UpperName	LowerName
☐	Edit	Copy	Delete	1	ALI KHAN	ali khan
☐	Edit	Copy	Delete	2	SARA IQBAL	sara iqbal
☐	Edit	Copy	Delete	3	HAMZA RAZA	hamza raza
☐	Edit	Copy	Delete	4	AYESHA NOOR	ayesha noor
☐	Edit	Copy	Delete	5	BILAL AHMED	bilal ahmed
☐	Edit	Copy	Delete	6	FATIMA SHEIKH	fatima sheikh
☐	Edit	Copy	Delete	7	USMAN TARIQ	usman tariq
☐	Edit	Copy	Delete	8	MAIRA JAVED	maira javed

Task A3 — Trimmed Name and Character Count

➤ Objective

For each customer, show the trimmed name and number of characters in it.

➤ SQL Query

```
SELECT
    CustID,
    TRIM(CustName) AS CleanedName,
    CHAR_LENGTH(TRIM(CustName)) AS NameLength
FROM Customer;
```

➤ Explanation

First, TRIM() removes unnecessary spaces. Then CHAR_LENGTH() counts the characters in the cleaned name.

➤ Result

The output contains:

- Customer ID
- Cleaned name
- Number of characters



The screenshot shows a data table with three columns: CustID, CleanedName, and NameLength. Each row represents a customer and includes interactive icons for editing, copying, and deleting. The table is displayed within a window that has a search bar at the top and a toolbar at the bottom.

		CustID	CleanedName	NameLength
☐	Edit	1	Ali Khan	8
☐	Edit	2	Sara Iqbal	10
☐	Edit	3	HAMZA RAZA	10
☐	Edit	4	Ayesha Noor	11
☐	Edit	5	bilal ahmed	11
☐	Edit	6	Fatima Sheikh	13
☐	Edit	7	Usman Tariq	11
☐	Edit	8	Maira Javed	11

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Task A4 — Build a Greeting

➤ Objective

Build a greeting for each customer in the following format:

Dear <trimmed name>, welcome!

➤ SQL Query

```
SELECT
    CustID,
    CONCAT('Dear ', TRIM(CustName), ', welcome!') AS Greeting
FROM Customer;
```

➤ Explanation

TRIM() cleans the customer name and CONCAT() combines the name with the greeting text.

➤ Example

Dear Ali Khan, welcome!

Extra options				
			CustID	Greeting
☐	Edit	Copy	Delete	1 Dear Ali Khan, welcome!
☐	Edit	Copy	Delete	2 Dear Sara Iqbal, welcome!
☐	Edit	Copy	Delete	3 Dear HAMZA RAZA, welcome!
☐	Edit	Copy	Delete	4 Dear Ayesha Noor, welcome!
☐	Edit	Copy	Delete	5 Dear bilal ahmed, welcome!
☐	Edit	Copy	Delete	6 Dear Fatima Sheikh, welcome!
☐	Edit	Copy	Delete	7 Dear Usman Tariq, welcome!
☐	Edit	Copy	Delete	8 Dear Maira Javed, welcome!
☐ Check all With selected: Edit Copy Delete Export				

Task A5 — Extract Email Username

➤ Objective

For customers who have an email address, extract the part before @.

➤ SQL Query

```
SELECT
  CustName,
  SUBSTRING(Email, 1, LOCATE('@', Email) - 1) AS Username
FROM Customer
WHERE Email IS NOT NULL;
```

➤ Explanation

Task A6 — Extract Email Domain

➤ Objective

For customers who have an email, extract everything after @.

➤ SQL Query

```
SELECT
  CustName,
  SUBSTRING(Email, LOCATE('@', Email) + 1) AS Domain
```

FROM Customer
WHERE Email IS NOT NULL;

➤ Explanation

LOCATE() finds the position of @, and SUBSTRING() extracts everything after it.

For example:

ali.khan@MAIL.com

produces:

MAIL.com

Extra options

←T→

▼

					CustName	Domain		
☐		Edit		Copy		Delete	Ali Khan	MAIL.com
☐		Edit		Copy		Delete	Sara Iqbal	example.com
☐		Edit		Copy		Delete	HAMZA RAZA	example.com
☐		Edit		Copy		Delete	bilal ahmed	MAIL.COM
☐		Edit		Copy		Delete	Fatima Sheikh	example.com
☐		Edit		Copy		Delete	Usman Tariq	example.com
☐		Edit		Copy		Delete	Maira Javed	example.com

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Task A7 — Display First Three Characters

➤ Objective

Show the first three characters of each customer's name.

➤ SQL Query

```
SELECT
    CustID,
    CustName,
    LEFT(TRIM(CustName), 3) AS First3Characters
FROM Customer;
```

➤ Explanation

LEFT() returns the first three characters of the cleaned customer name.

Extra options

← T →				CustID	CustName	First3Characters
☐	 Edit	 Copy	 Delete	1	Ali Khan	Ali
☐	 Edit	 Copy	 Delete	2	Sara Iqbal	Sar
☐	 Edit	 Copy	 Delete	3	HAMZA RAZA	HAM
☐	 Edit	 Copy	 Delete	4	Ayesha Noor	Aye
☐	 Edit	 Copy	 Delete	5	bilal ahmed	bil
☐	 Edit	 Copy	 Delete	6	Fatima Sheikh	Fat
☐	 Edit	 Copy	 Delete	7	Usman Tariq	Usm
☐	 Edit	 Copy	 Delete	8	Maira Javed	Mai



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 Delete

 Export

Task A8 — Mask Phone Numbers

➤ Objective

Show the first four characters of each phone number followed by:

XXX-XXXX

Customers without a phone number should be skipped.

➤ SQL Query

```
SELECT
    CustName,
    Phone,
    CONCAT(LEFT(Phone, 4), 'XXX-XXXX') AS MaskedPhone
FROM Customer
WHERE Phone IS NOT NULL;
```

➤ Explanation

LEFT() extracts the first four characters, while CONCAT() combines them with the masking text.

➤ Example

0300-1112233

becomes:

0300XXX-XXXX

Extra options

				CustName	Phone	MaskedPhone
☐				Ali Khan	0300-1112233	0300XXX-XXXX
☐				Sara Iqbal	0301-4445566	0301XXX-XXXX
☐				HAMZA RAZA	0302-7778899	0302XXX-XXXX
☐				Ayesha Noor	0303-1234567	0303XXX-XXXX
☐				bilal ahmed	0304-2345678	0304XXX-XXXX
☐				Fatima Sheikh	0305-3456789	0305XXX-XXXX
☐				Maira Javed	0307-5678901	0307XXX-XXXX

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Task A9 — Replace Spaces with Hyphens

➤ Objective

Show each product name with all spaces replaced by hyphens.

➤ SQL Query

```
SELECT
    ProdID,
    ProdName,
    REPLACE(ProdName, ' ', '-') AS SlugName
FROM Product;
```

➤ Explanation

REPLACE() changes every space into a hyphen.

➤ Example

Laptop Pro 15

becomes:

Laptop-Pro-15

				ProdID	ProdName	SlugName
☐	Edit	Copy	Delete	101	Laptop Pro 15	Laptop-Pro-15
☐	Edit	Copy	Delete	102	Wireless Mouse	Wireless-Mouse
☐	Edit	Copy	Delete	103	USB-C Cable	USB-C-Cable
☐	Edit	Copy	Delete	104	Office Chair	Office-Chair
☐	Edit	Copy	Delete	105	Standing Desk	Standing-Desk
☐	Edit	Copy	Delete	106	Notebook A4	Notebook-A4
☐	Edit	Copy	Delete	107	Ballpoint Pen 10pk	Ballpoint-Pen-10pk
☐	Edit	Copy	Delete	108	Coffee Beans 1kg	Coffee-Beans-1kg
☐	Edit	Copy	Delete	109	Green Tea Box	Green-Tea-Box
☐	Edit	Copy	Delete	110	Bluetooth Speaker	Bluetooth-Speaker

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Task A10 — Pad Product IDs

➤ Objective

Display each product ID padded to five digits using leading zeros.

➤ SQL Query

```
SELECT
  ProdID,
  LPAD(ProdID, 5, '0') AS PaddedID
FROM Product;
```

➤ Explanation

LPAD() adds zeros to the left side until the value contains five characters.

➤ Example

101

becomes:

00101

				ProdID	PaddedID
☐	 Edit	 Copy	 Delete	101	00101
☐	 Edit	 Copy	 Delete	102	00102
☐	 Edit	 Copy	 Delete	103	00103
☐	 Edit	 Copy	 Delete	104	00104
☐	 Edit	 Copy	 Delete	105	00105
☐	 Edit	 Copy	 Delete	106	00106
☐	 Edit	 Copy	 Delete	107	00107
☐	 Edit	 Copy	 Delete	108	00108
☐	 Edit	 Copy	 Delete	109	00109
☐	 Edit	 Copy	 Delete	110	00110

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Task A11 — Find Products Containing "Pro"

➤ Objective

Show product names that contain the word Pro and display the position of Pro.

➤ SQL Query

```

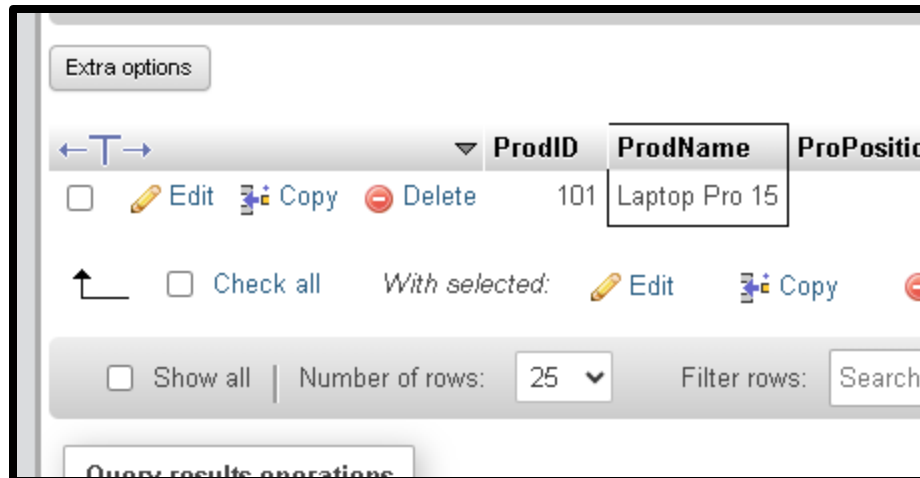
SELECT
  ProdID,
  ProdName,
  LOCATE('Pro', ProdName) AS ProPosition
FROM Product
WHERE LOCATE('Pro', ProdName) > 0;

```

➤ Explanation

LOCATE() searches for Pro inside the product name and returns its starting position.

The WHERE condition ensures that only products containing Pro are displayed.



Task A12 — Extract Customer First Name

➤ Objective

Display every customer's first name only.

➤ SQL Query

```
SELECT
    CustID,
    CustName,
    SUBSTRING(
        TRIM(CustName),
        1,
        LOCATE(' ', TRIM(CustName)) - 1
    ) AS FirstName
FROM Customer;
```

➤ Explanation

The query performs three operations:

1. TRIM() removes unnecessary spaces.
2. LOCATE() finds the first space between the first and last name.
3. SUBSTRING() extracts the text before that space.

Extra options

				CustID	CustName	FirstName
☐		Edit		Copy		Delete
☐		Edit		Copy		Delete
☐		Edit		Copy		Delete
☐		Edit		Copy		Delete
☐		Edit		Copy		Delete
☐		Edit		Copy		Delete
☐		Edit		Copy		Delete
☐		Edit		Copy		Delete


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7. Conclusion

Part A of the Scalar SQL Functions lab demonstrated the use of different MySQL string functions. The functions were used to clean customer names, change text case, count characters, create greetings, extract email usernames and domains, extract parts of names, mask phone numbers, modify product names, format product IDs, and locate specific text.

The lab showed that scalar functions are useful for transforming raw database data into clean and human-readable information.
